

The Alan Machine

João Alisson and contributors

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Preface

For Alan Turing, who imagined the machine,
and for Alan Ricardo.

How to read this book

This book is about a machine that does not exist. We call it Alan, and we use it to ask a question physics takes very seriously: can entropy be reversed, and if it could, what would happen to time?

You do not need a physics degree to read it. The chapters are written for anyone who is curious. When the math helps, it shows up in boxes marked *Going deeper*, which you can open or skip. The story works either way.

Some chapters have a companion paper, written for specialists, with the full argument. You will find the papers in the last part of the book, and each chapter tells you when one exists.

Along the way, we keep three kinds of statement apart. Unmarked claims are **established** physics, with references you can check. Boxes marked **Extrapolation** follow from established physics, but only if their stated assumptions hold. Boxes marked **Speculation** hold ideas that are conceivable but that no evidence supports yet. A thought experiment is only useful if you always know which of the three you are reading.

Part I

The Machine and the Question

1 Meet Alan

This chapter has not been written yet.

2 What Entropy Is

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3 The Arrow of Time

This chapter has not been written yet.

Part II

Information Is Physical

4 Maxwell's Demon and the Szilard Engine

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5 Landauer's Principle

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6 Reversible Computing

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Part III

The Limits of the Machine

7 How Fast Can It Compute

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8 How Much Can It Hold

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9 Black Holes and Information

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Part IV

Reversing Entropy

10 Local Reversal

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Part V

The Impact on Time

13 Would Time Run Backward

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14 Closed Timelike Curves

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15 The End of the Universe

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Epilogue: What Alan Cannot Do

This chapter has not been written yet.

Part VI

Papers

About the Papers

The chapters tell the story. The papers make the argument.

Some chapters have a companion paper with the full scientific treatment of their subject: the derivations, the assumptions and the references a specialist would expect. Each paper stands on its own, with its own authors, abstract and references, and can be cited independently of the book. A chapter can lean on several papers, and a paper can support several chapters.

No papers have been written yet.

References

A Glossary

This appendix has not been written yet.

B Notation and Units

This appendix has not been written yet.

C Physical Constants

This appendix has not been written yet.

D A Short Primer on Statistical Mechanics

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